

Stratum AI

AI Incident Copilot for Enterprise Operations

"From searching documentation to resolving incidents."

Team Elite Party · HPE Synergy 2026

The Hidden Cost of Incident Response

Every production incident triggers a fragmented, manual search across siloed knowledge systems — costing engineers hours and businesses revenue. The tools exist, but they don't talk to each other — and engineers pay the price under pressure.

Documentation Sprawl

Runbooks on GitHub, wikis on Confluence, cloud docs on AWS/GCP/Azure — no single source of truth. Engineers waste critical minutes locating the right document before they can even begin diagnosis.

Context Switching Tax

Engineers toggle between 4–6 tools during a live incident, compounding cognitive load under pressure. Every context switch is a window for errors and delays in a high-stakes environment.

MTTR Impact

Static documentation cannot adapt to incident context — every search starts from zero. There is no institutional memory available at the moment it matters most.

Business Downtime

Delayed resolution directly translates to SLA breaches, revenue loss, and customer trust erosion. The cost compounds with every minute an incident remains unresolved.

Incident Response Breakdown



Each stage in the fragmented workflow adds latency and cognitive overhead. The final stage — Business Downtime — is the avoidable consequence of a broken knowledge retrieval process.

Introducing **Stratum AI**

AI-Powered Incident Copilot — one intelligent interface that replaces fragmented search with grounded, context-aware operational guidance. Stratum AI is built for production, not demos.

1

Enterprise RAG Engine

Retrieval-Augmented Generation over your actual runbooks, GitHub repos, and cloud documentation — no generic knowledge base.

2

Hybrid Retrieval

Combines dense vector search (FAISS) with sparse BM25 — best of semantic and keyword matching for maximum recall precision.

3

Context-Aware Assistance

Understands incident context to surface the most relevant remediation steps — not just keyword matches from a flat document index.

4

Grounded Responses + Citations

Every answer is traceable to a source document. No hallucinated guidance in production — confidence scoring on every response.

5

Interactive Troubleshooting

Conversational interface guides engineers step-by-step through resolution workflows — from alert to remediation path in seconds.

AI Chat Panel

Natural language incident queries answered with structured, grounded guidance. Ask in plain English — receive actionable steps.

Source Citations Sidebar

Every response links directly to the originating document — runbook, GitHub file, or cloud doc. Full traceability for every claim.

Confidence Score Meter

Inline confidence indicator lets engineers calibrate trust in AI guidance. High-confidence answers are flagged as production-safe.

Recommended Actions Panel

Ranked remediation suggestions surfaced from the knowledge corpus, ordered by relevance to the active incident context.

System Architecture

A fully local, enterprise-grade RAG pipeline — from raw documentation to grounded AI responses, with **zero external API dependencies**. Every component runs on-premise, inside your security perimeter.



01

Ingestion Layer

Runbooks, GitHub repos, cloud docs, and internal knowledge bases are parsed and normalized. Structure-aware chunking preserves document semantics and hierarchy.

02

Embedding & Dual Indexing

Chunks are embedded into dense vectors and stored in FAISS. Simultaneously indexed in BM25 for keyword retrieval. Two indices, one unified query path.

03

Hybrid Retrieval Engine

Reciprocal Rank Fusion merges semantic and keyword result sets — maximizing recall while preserving precision for ambiguous incident queries.

01

Local LLM Generation

Retrieved context is passed to an on-premise language model. The response generator injects citations and confidence scores before delivery — no cloud roundtrips.

02

Engineer-Facing Dashboard

Clean web UI surfaces AI guidance, source citations, confidence indicators, and recommended actions — purpose-built for incident response workflows.

03

Zero External APIs

The entire pipeline — embedding, indexing, retrieval, and generation — runs locally. Your runbooks and incident data never leave your infrastructure.

What Makes **Stratum AI** Different

Not another wrapper around a public LLM. A purpose-built, enterprise-safe AI copilot with architectural decisions that matter in production. The differentiation is structural — not cosmetic.

Capability	Traditional Search	Stratum AI
Retrieval Method	Keyword Search	Hybrid: Semantic + BM25
Understanding	Literal Match	Semantic Context Awareness
Response Quality	Raw Document Links	Grounded AI with Citations
Hallucination Risk	N/A	Minimized via RAG + Confidence Scoring
Resolution Speed	Slow — Manual Reading	Fast — Interactive Guidance
Data Privacy	Varies	100% Local LLM, Zero External APIs
Documentation Type	Static	Adaptive to Incident Context

📌 Key differentiators — **Local LLM**, **Zero External APIs**, and **Confidence Scoring** — are architectural commitments, not feature flags. They are built into the foundation of Stratum AI from day one.

Current Development Progress

Core RAG pipeline is **operational**. Enterprise integrations and analytics layer are next. The foundation is production-validated — the roadmap builds on solid ground.

✓ Completed Milestones

- **Repository & Infrastructure**
Project initialized, architecture finalized, dev environment fully configured and version-controlled.
- **Hybrid Retrieval**
FAISS vector index + BM25 search implemented, integrated, and validated against test document corpora.
- **Document Processing**
Markdown parser with structure-aware chunking operational — preserves heading hierarchy and code block semantics.
- **Citation Engine**
Source attribution and confidence scoring integrated into the response generation pipeline end-to-end.
- **Local LLM Integration**
On-premise model connected to retrieval pipeline. Fully air-gapped — zero external API calls at generation time.
- **Console UI**
Functional CLI interface operational for testing, validation, and pipeline debugging during development.

☐ Next Phase Targets

- Enterprise Integrations**

Confluence, Jira, and PagerDuty connectors — bidirectional data flow from incident management systems directly into the ingestion layer.
- Incident Analytics & RBAC**

Role-based access control, per-incident feedback loop, MTTR trend dashboards, and audit-ready monitoring — the enterprise control plane.

Pipeline Status



- Overall Completion**

Core platform complete.
Integration layer in progress.
- RAG Pipeline**

Retrieval, generation, and citation engine fully operational.

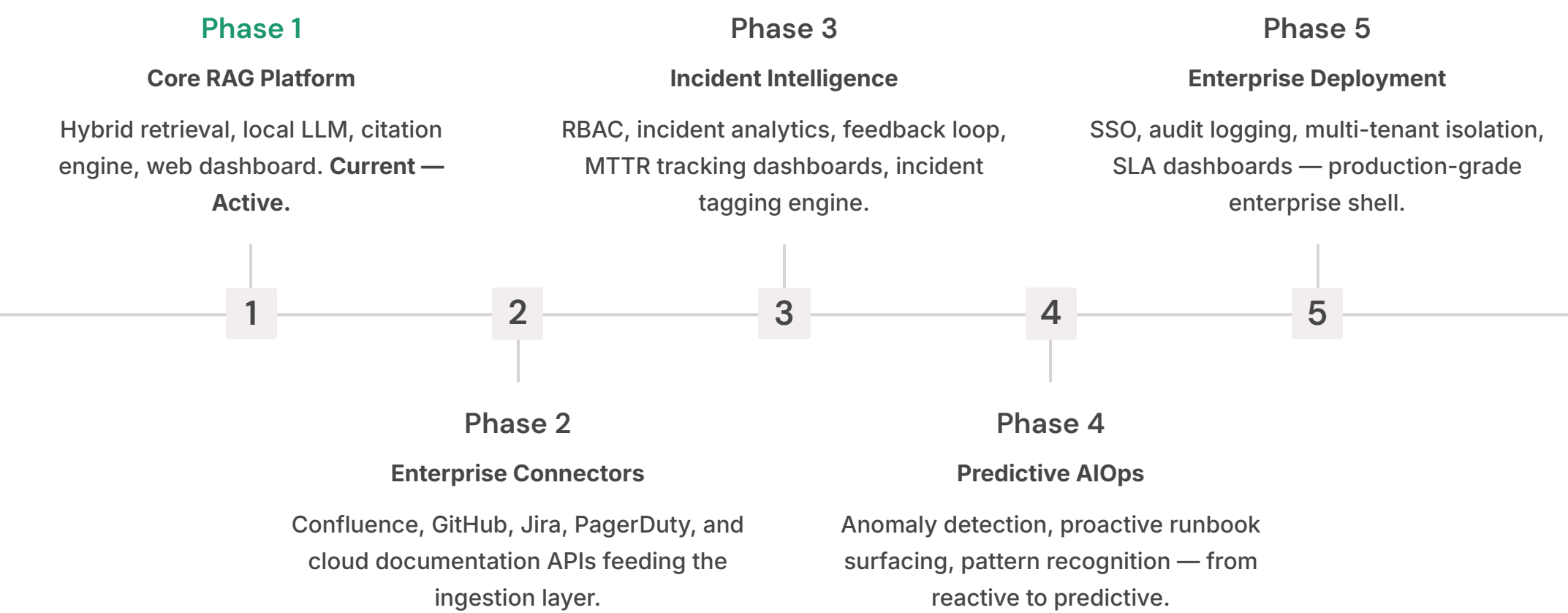


Enterprise Connectors

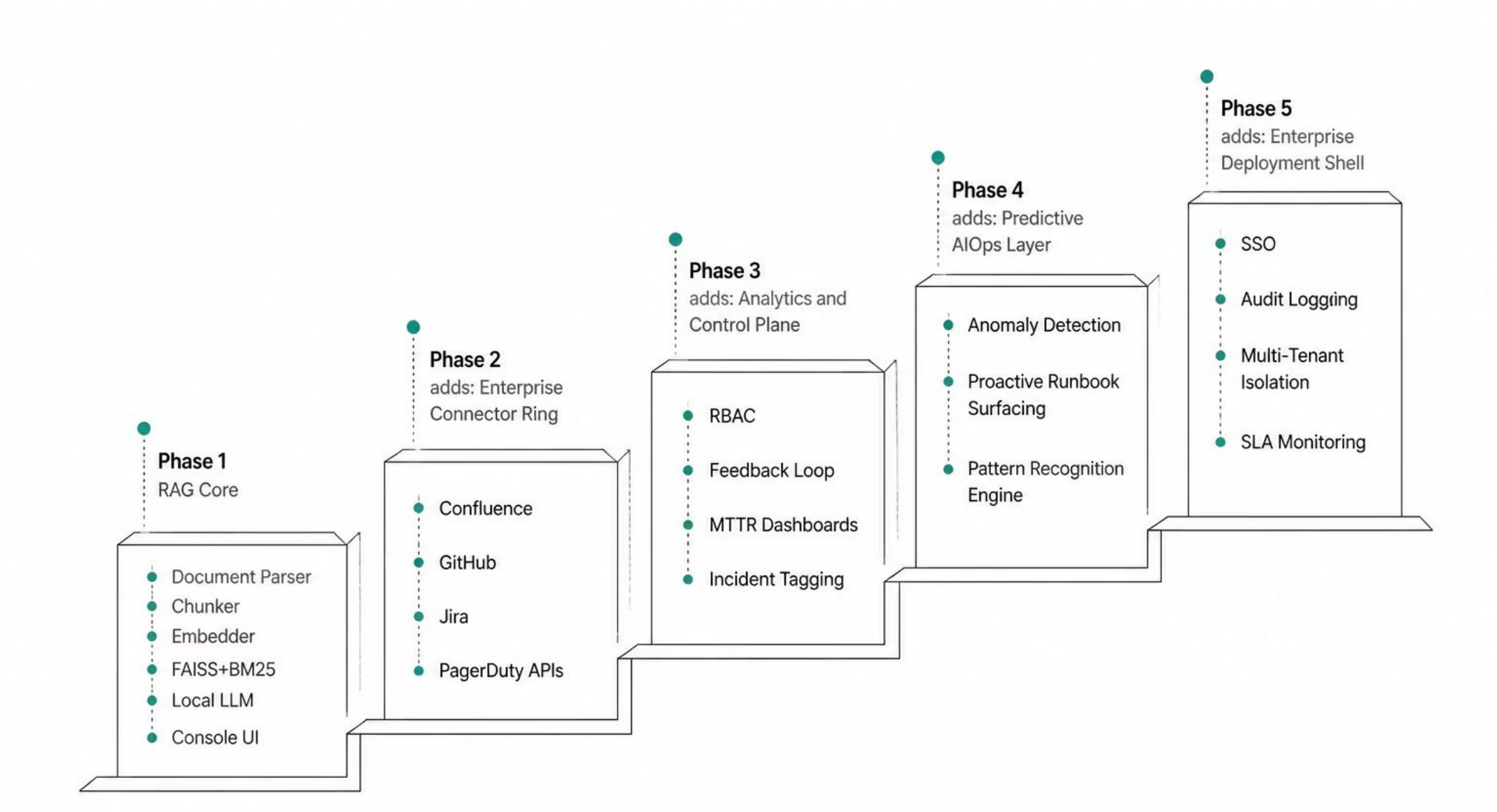
- Architecture finalized.
Implementation next sprint.

Roadmap to Enterprise Deployment

Five phases from core platform to full AIOps intelligence — each phase adds a new architectural layer on a production-validated foundation. Phase 1 is live. The path to enterprise scale is defined.



Architecture Evolution — Stack Growth Per Phase



Each phase is additive — the architecture grows upward without replacing prior layers. Phase 1 forms the immutable core that every subsequent capability depends on. No phase requires a rearchitecture of what came before.

Beyond a Chatbot

"We are not building another AI chatbot. We are building an AI Incident Copilot that empowers engineers with trusted operational intelligence, accelerates incident resolution, and transforms enterprise knowledge into actionable guidance."

Trusted

Every response grounded in your documentation. Every claim cited to a source. No hallucinated guidance reaches production.

Fast

From alert to resolution path in seconds, not hours. Context-aware retrieval eliminates the manual search tax entirely.

Private

Fully local — your runbooks never leave your infrastructure. Zero external API calls. Complete data sovereignty by design.

Scalable

Built for enterprise from day one — not retrofitted. Architecture designed to grow from core RAG to full AIOps intelligence.

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Thank You

"From searching documentation to resolving incidents."

We are now open for questions!

Team

Team Elite Party

Hackathon

HPE Synergy 2026

Project

Stratum AI

Category

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